

Region-Grounded Fine-Tuning for Late-Interaction VLMs

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Project Proposal

1 Motivation and Preliminary Literature Survey

Vision-Language Models (VLMs) such as CLIP [1] and SigLIP [2] have achieved remarkable success in zero-shot image classification and cross-modal retrieval. However, these models inherently align whole images to whole captions using a global pooling mechanism. This creates representations with weak spatial structure and poor compositional understanding—often failing to distinguish between “a horse eating grass” and “grass eating a horse” [7], or struggling with dense prediction tasks like zero-shot semantic segmentation.

Late-interaction models such as FILIP [3] and FLAIR [4] attempt to resolve this by matching at the patch-token and word-token levels. Yet, even in these architectures, the alignment is learned implicitly from global image–caption pairs; there is no explicit supervision telling the model which specific word corresponds to which spatial patch. Conversely, models that utilize explicit region supervision often rely on expensive human-annotated bounding boxes (e.g., RefCOCO or Visual Genome).

Recent work in mechanistic interpretability and attention mapping, such as SCLIP [5] and MaskCLIP [6], has demonstrated that CLIP-family models inherently encode rough spatial structure that can be extracted without fine-tuning. This project aims to bridge these two paradigms. We propose generating region-level semantic supervision automatically—without human annotation—by utilizing frozen VLMs to extract crops and caption them, thereby injecting strong spatial grounding into a base model via a modified late-interaction loss.

2 Main Questions and Goals

The primary hypothesis of this project is that **region-level supervision, self-generated via an ensemble of frozen vision and language models, can significantly improve fine-grained compositional understanding and dense visual grounding in late-interaction VLMs, without degrading their global zero-shot performance.**

If successful, the outcomes of this project will be twofold:

1. **A Novel Supervision Pipeline:** An automated pipeline for generating highly localized, noise-filtered (region, caption) pairs from standard global datasets (e.g., CC3M).
2. **Enhanced Model Capabilities:** A fine-tuned SigLIP checkpoint equipped with dense spatial grounding capabilities, improving its utility for downstream dense prediction tasks and compositional reasoning.

Even a negative result (e.g., failing to improve over the baseline) will be highly informative, as it will establish empirical bounds on how much fine-grained information can be recovered purely from pseudo-labels generated by CLIP-extracted spatial clusters.

3 Tentative Plan and Methodology

The project consists of three distinct stages, moving from data generation to model fine-tuning.

3.1 Stage 1: Spatial Region Extraction

We will extract a handful of spatially coherent regions per image. Utilizing a frozen base SigLIP model, we will apply an approach similar to SCLIP’s correlation attention. By analyzing the self-attention maps of the final transformer layers, we can cluster semantically consistent patches. For a given image I , this process yields a set of cropped regions $R = \{r_1, r_2, \dots, r_k\}$.

3.2 Stage 2: Auto-Captioning and Filtering

Each extracted region r_i will be passed through a small, efficient Vision-Language Model (e.g., Qwen2-VL-2B) to generate a localized caption c_i . Because VLMs are prone to hallucination (especially on tightly cropped, low-context patches), we will introduce a filtering step. We will compute the zero-shot SigLIP similarity score between the crop r_i and the generated caption c_i . If the similarity $s(r_i, c_i)$ falls below a threshold τ , the pair is discarded. This yields a cleaned dataset of localized pairs $\mathcal{D}_{region} = \{(r_i, c_i)\}$.

3.3 Stage 3: Model Fine-Tuning Protocol

We will perform Low-Rank Adaptation (LoRA) fine-tuning on the attention and MLP weights of SigLIP-B/16. The training objective will be a combination of the standard global loss and a late-interaction region loss. Let the standard global SigLIP pairwise sigmoid loss be $\mathcal{L}_{global}$. For the region-level supervision, we adapt the FILIP token-wise maximum similarity loss $\mathcal{L}_{late}$. For image patch tokens $v_k \in \mathbb{R}^d$ and text word tokens $w_m \in \mathbb{R}^d$, the cross-modal similarity is computed as:

$$s(r_i, c_i) = \frac{1}{|c_i|} \sum_m \max_k (v_k^\top w_m)$$

The region loss $\mathcal{L}_{region}$ is then formulated using this late-interaction similarity. The total loss for fine-tuning will be:

$$\mathcal{L}_{total} = \mathcal{L}_{global} + \lambda \mathcal{L}_{region}$$

where λ is a hyperparameter balancing global alignment and local grounding.

4 Measures of Success and Evaluation Design

To rigorously evaluate the impact of our self-generated region supervision, we will measure success across three distinct axes. We will compare our model against two baselines: (1) Off-the-shelf SigLIP-B/16, and (2) SigLIP-B/16 fine-tuned purely on the global loss over the same dataset (to control for the effect of the fine-tuning process itself).

4.1 Zero-Shot Global Alignment (Guardrail)

To ensure that region-level fine-tuning does not destroy the model’s global understanding, we will measure zero-shot Top-1 classification accuracy on ImageNet-1K and CIFAR-100. The goal here is parity with the baseline.

4.2 Dense Grounding (Zero-Shot Segmentation)

We will extract the patch-level representations from the fine-tuned vision encoder and evaluate zero-shot semantic segmentation on **PASCAL VOC** and **COCO-Stuff**. Success will be measured by the mean Intersection over Union (mIoU). We hypothesize a strict numerical improvement over the baselines due to the explicit patch-word alignment learned via $\mathcal{L}_{late}$.

4.3 Compositionality

To test if the model actually understands relationships and binding, we will evaluate on **ARO** (Attribution, Relation, and Order), **SugarCrepe**, and **Winoground**. These benchmarks measure the model’s ability to distinguish a caption from heavily confounded negative counterparts (e.g., swapped word order).

5 Data, Compute, and Feasibility

Data: We will train on a 50k–100k random subset of the Conceptual Captions 3M (CC3M) dataset. The region-caption triples will be generated in-house using the Stage 1 and 2 pipeline. All evaluation benchmarks are public and readily accessible.

Compute Requirements: The compute budget targets Colab Pro using A100 spot instances (falling back to L4 if necessary).

- *Preprocessing:* 20–30 A100-hours to generate the region captions. Running Qwen2-VL-2B inference is the primary bottleneck, but it is a one-time upfront cost.
- *Training:* 15–20 A100-hours per fine-tuning run. We budget for approximately 5 runs to account for hyperparameter tuning (e.g., the threshold τ and loss weight λ) and ablation studies.

Feasibility: If A100 access proves too spotty, we will seamlessly scale down to a SigLIP-B/32 backbone on a 30k CC3M subset. Given the focused scope of the LoRA fine-tuning and the one-pass nature of the data generation, this project is highly feasible to complete within the 5-week course timeframe.

Note : Combined together we have about 300 credits on Google Colab

6 Stretch Goals

If time permits, we plan to explore two extensions:

- **Bootstrapping the Loop:** Feed the fine-tuned model back into Stage 1 to generate superior correlation attention maps, generating higher-quality regions for a second pass of VLM captioning, testing if the performance compounds or saturates.
- **FLAIR Backbone Integration:** Swap the SigLIP backbone for FLAIR, provided its released checkpoint is mature. FLAIR’s mechanism of attention-pooling over text queries inherently fits region-level supervision more naturally than SigLIP’s global pooling formulation.

7 Work Division:

We will both be doing everything together

References

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- [5] Zifeng Wang, et al. SCLIP: Rethinking Self-Attention in CLIP for Zero-shot Semantic Segmentation. *arXiv*, 2023.
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